Mars-JEPA: Multispectral Joint Embedding Predictive Architectures for Martian Landslide Segmentation

BMVC 2026 Submission # 1133

Abstract

The expansion of autonomous exploration on Mars has unlocked unprecedented scientific frontiers, yet it requires navigating a landscape characterized by frequent landslides that drive geomorphological change by mass movement of rock and debris. This presents critical mechanical hazards to robotic assets. Mitigating these risks through accurate landslide prediction is therefore essential to safeguarding mission longevity and preventing catastrophic loss. While landslide prediction from multispectral images using traditional methods yields high accuracy, its generalizability is limited. In this work, we study the performance of self-supervised learning-based (SSL) image models, specifically trained as joint-embedding predictive architectures (JEPA) and develop a novel physics-aware image encoder, showing additional performance gains while maintaining the generalizability provided by SSL models. Experiments reveal that our physics-aware masking strategy captures long-range dependencies between spectral bands. We show that by learning cross-spectral latent correlations rather than raw pixel reconstructions, our model is significantly more robust to the sensor noise and varying lighting conditions commonly encountered in multispectral remote sensing data. We train JEPA encoders employing different masking strategies, and benchmark their downstream landslide segmentation performance on the MMLSv2 dataset. This work directly improves autonomous safety for planetary exploration and establishes a scalable framework for self-supervised learning in high-dimensional remote sensing and terrestrial disaster response.

1 Introduction

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2 Related Work

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2.1 Planetary Surface Analysis and Exploration

Mars has long been a focal point for planetary exploration, with significant research dedicated to identifying landslides on its surface. Detecting landslides automatically is a non-trivial task due to shifting atmospheric conditions and complex geomorphology. Historically, experts have identified these features manually using high-resolution imagery from the High Resolution Imaging Science Experiment (HiRISE) [5].

Early automated systems primarily utilized physics-based features derived from digital elevation models, such as slope and elevation. Recently, the field has transitioned toward deep learning, employing Convolutional Neural Networks (CNNs) [2] and architectures like U-Net [7] for automated mapping. While traditional studies relied on monochromatic imagery, the introduction of the MMLSv2 dataset [6] enabled the use of multimodal data, combining spectral imagery with topographic information to improve detection accuracy.

2.2 Self-Supervised Learning in Remote Sensing

Self-supervised representation learning has seen a paradigm shift from contrastive frameworks, which rely heavily on data augmentations, to Masked Image Modeling (MIM). While generative approaches like Masked Autoencoders (MAE) [3] have shown promising results by reconstructing raw pixels, they are often ill-suited for the multi-sensor nature of Martian remote sensing. Pixel reconstruction is computationally intensive for high-dimensional data and risks "information leakage" in multimodal datasets, where highly correlated channels (e.g., RGB and Grayscale) allow the model to bypass learning deeper geological features.

2.3 Joint Embedding Predictive Architectures (JEPA)

At the core of our methodology is the JEPA architecture, a paradigm proposed as a foundational step toward non-generative world models [4]. Unlike generative models, JEPA performs its predictive objective entirely within an abstract representation space. Following the I-JEPA approach [1], our model utilizes an asymmetrical structure consisting of a context encoder, a target encoder, and a predictor.

The context encoder processes only visible "hints" of the Martian surface, while the predictor estimates the latent embeddings of missing patches. This "prediction in latent space" forces the network to capture underlying semantic and geological properties rather than superficial spectral values. To prevent representation collapse—a common challenge in self-supervised learning—we integrate Stochastic Information Gradient Regularization (SIGReg), ensuring that learned features remain diverse across all seven input modalities.

2.4 Spatio-Spectral Reasoning

Our work extends the JEPA framework into the spatio-spectral domain. Unlike previous implementations that treat multimodal inputs as a single unified cube, our architecture utilizes independent channel tokenization and factorized learned embeddings. This allows for a granular masking strategy where specific modalities can be hidden at specific spatial locations. By doing so, we force the model to learn the intrinsic physical correlations between disparate Martian sensors—such as the relationship between topographic slope and thermal inertia—thereby producing more robust features for landslide segmentation.

2.5 Semantic Segmentation

Write about modern semantic segmentation heads and how they have been used with different encoder backbones.

3 Methodology

In this work, we study the downstream performance on segementation tasks for two JEPAs trained with different tokenization masking strategies, on multispectral images. This section details the formulation and development of the self-supervised models for segmentation as follows: Section 3.1 formulates the landslide segmentation problem on multispectral images; Section 3.2 provides an overview of the general architecture and methodology to train a JEPA encoder; Section 3.3 discusses the adaptation of the I-JEPA [**?**] architecture for multispectral images; Section 3.4 provides a novel spatial-spectral masking strategy for training JEPAs on multispectral images, and finally Section 3.5 discusses the integration of the JEPA encoders with SOTA segmentation architectures for landslide segmentation on the Martian surface.

3.1 Problem Formulation

Consider a multispectral image $I \in \mathbb{R}^{C\times H\times W}$, where C is the number of bands, H,W are the height and width of the image respectively. We define a binary segmentation mask $M \in \{0,1\}^{H\times W}$ over the pixels in the image, where a pixel $M[i,j]$ at row i and column j in the segmentation mask M has a value of 1 if it is predicted to be affected by a landslide, and 0 otherwise.

The Martian landslide detection problem is to develop a segmentation model $S_\theta : \mathbb{R}^{C\times H\times W} \to \{0,1\}^{H\times W}$, parameterized by θ, to satisfy the objective

$$\max_{\theta} \mathbb{E}_{(I,M)\sim D}\left[\sigma_{\text{sim}}(S_\theta(I), M)\right], \tag{1}$$

where the multispectral image and corresponding segmentation mask pairs (I,M) are sampled from the dataset D, and $\sigma_{\text{sim}}(M_1,M_2) \in \mathbb{R}$ is a similarity metric between two binary segmentation masks M_1 and M_2. Concretely, the objective aims to maximize the expected similarity between the segmentation mask predicted by the model and the actual segmentation mask, for all examples in the dataset, by performing an optimization over the model parameters θ.

3.2 Joint Embedding Predictive Architectures

3.3 Adapting I-JEPA for Multispectral Image Encoding

3.4 Spatio-Spectral JEPA

3.5 Segmentation Models with Self-Supervised Backbones

4 Experimental Setup

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5 Results and Discussion

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Encoder	Segmentation	MIoU	Precision	Recall	F1 Score
I-JEPA	UperNet	0.8350	0.8613	0.8678	0.8643
	UNet	0.8286	0.8531	0.8723	0.8624
	UNet++	0.8247	0.8601	0.8795	0.8690
	FPN	0.8246	0.8488	0.8897	0.8687
	MANet	0.8100	0.8448	0.8683	0.8555
	DeepLabv3+	0.8183	0.8576	0.8689	0.8632
	PAN	0.8141	0.8559	0.8693	0.8622
SS-JEPA	UperNet	**0.8652**	0.8838	0.8859	0.8848
	UNet	0.8521	0.8631	0.8868	0.8748
	UNet++	0.8399	0.8768	0.8596	0.8681
	FPN	0.8265	0.8656	0.8777	0.8716
	MANet	0.8262	0.8691	0.8703	0.8697
	DeepLabv3+	0.8202	0.8579	0.8660	0.8619
	PAN	0.8170	0.8620	0.8587	0.8604

Table 1: Segmentation performance with self-supervised encoder backbones.

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6 Conclusion

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